

In-Context Robot Learning with General Agents

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Humans continually adapt to new situations, whereas a robot trained on a finite set of demonstrations must face tasks and environments that training cannot exhaustively cover. We ask whether a fixed commercial language or vision-language model (LLM/VLM) can extract new task information from a small test-time context and turn it into executable, verifiable behavior from a new initial state. We define robotic in-context learning as adaptation during the target task that uses demonstrations, examples, or interaction feedback while forbidding gradient updates and persistent task-specific parameter changes. We introduce GPT-POLICY, a general-agent framework that accepts human videos, robot demonstrations, goal images, self-interaction history, and experience of environmental rules or human intent through a shared closed-loop interface for manipulation and exploration. A context compiler preserves task-relevant visual transitions, the VLM proposes robot-tool actions, and a constrained controller checks, executes, and reports each action. Initial task records and case analyses illustrate goal specification, changes in operation order, and online coordination, while exposing a gap between task reasoning and verified physical execution. The evaluation combines task-level success and efficiency reporting with matched model comparisons and planned context ablations to test the reliability and limits of these behaviors.

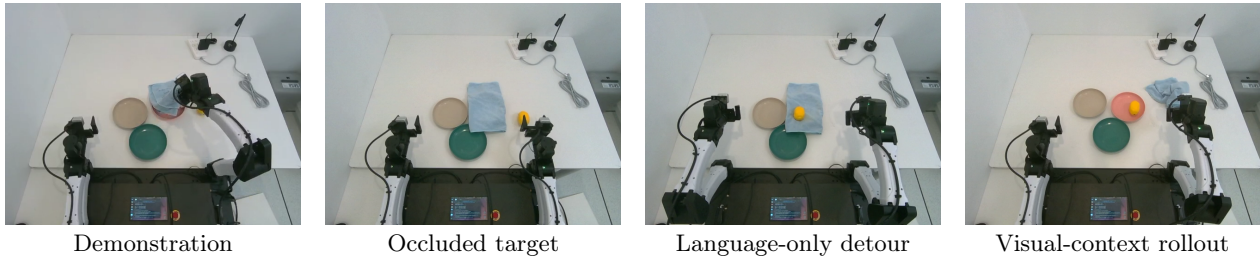


Figure 1 A visual demonstration can specify procedural structure that is absent from language. In the motivating example, the target plate is initially hidden by a towel. The language-only policy succeeds after an extended detour, whereas the visually conditioned policy follows the demonstrated strategy of uncovering the target before placement. We use controlled trials to test whether this behavior generalizes beyond the illustrated rollout.

1 Introduction

Humans continually adapt to unfamiliar situations by observing others, interpreting examples, and learning from the consequences of their own actions. Robots face the same need: a finite training distribution cannot enumerate every task, object arrangement, or interaction that deployment will bring. Test-time adaptation is therefore a key step toward a general embodied agent. Studying how much of this ability is already present in general language and vision-language models can help clarify which capabilities future embodied foundation models must learn explicitly and which can be elicited through context.

A brief task instruction may specify a desired outcome while leaving some execution details unstated. For example, the instruction *fold the towel* does not specify grasp points or a folding sequence; a demonstration can illustrate one way to carry out the task. In other situations, the agent may need information about the environment: placing a lemon on a plate that is currently occluded may require locating the plate from earlier observations or a demonstration. These examples motivate studying how agents use context to obtain task-relevant information beyond the immediate instruction and observation.

Recent robot foundation models show that a demonstration can condition a policy without task-specific

gradient updates. GEN-1.5 reports one-shot skill adaptation from physical prompts, including human-to-robot and sim-to-real examples (Generalist Team, 2026). S1 uses video demonstrations to specify novel atomic and long-horizon tasks (Skild AI, 2026), while Zero-WAM trains a video-action model to follow human video guidance on unseen tasks (Zhou et al., 2026). These systems are compelling, but they leave an important complementary question: *how much robotic in-context learning is already accessible through general VLMs that were not exposed as dedicated robot policies?*

Our central question is whether a fixed commercial LLM or VLM can extract new task information from a small number of demonstrations or interaction experiences at test time, transfer that information to a new manipulation situation, and generate decisions that can be executed and verified. We define robotic in-context learning as adaptation during the target task through demonstrations, examples, or interaction feedback that reveal task goals, object relations, procedures, or environmental rules. The model receives task-relevant context and performs no gradient updates or persistent task-specific parameter changes. Adaptation must be expressed in behavior from a new initial state, rather than assessed only through a verbal account of the example.

The context need not be a single demonstration video. We organize the study into five families: human demonstrations without a textual description, robot demonstrations with aligned actions, target images, the agent’s own interaction history, and experience revealing changing environmental rules or human intent. These families separate procedural transfer, fine contact, goal specification, exploration and correction, and interactive adaptation. Each family contains multiple constituent tasks, allowing the evaluation to distinguish the source of information from the particular task used to test it.

We investigate this question through GPT-POLICY, which uses a general VLM as a high-level closed-loop policy. The model observes live views, reasons about task progress, and invokes parameterized robot motion tools. In the human-video condition, the demonstration is represented by ordered visual keyframes without expert actions or privileged state. Other conditions provide a goal image, aligned robot actions, or accumulated interaction history through the same agent interface. A shared constrained execution layer translates the agent’s targets into robot motion and returns observations and outcomes for replanning. This makes it possible to compare context sources while controlling the action interface.

We report initial real-robot results and analyze how context affects goal specification, operation order, contact execution, exploration, and online coordination. The case studies suggest that context can help organize behavior while leaving a gap between a plausible plan and verified physical success. This motivates task-level model and context comparisons, together with ablations of action references, prompt detail, visual inputs, and retained history. The resulting analysis aims to identify when additional task information improves execution and when the remaining difficulty lies in perception, memory, or control.

2 Related Work

In-context learning for robot policies. In-context learning studies how models infer tasks and adapt their behavior from examples (Brown et al., 2020), with robotic antecedents in one-shot imitation and meta-learning (Duan et al., 2017). Existing approaches condition robot policies on multimodal task prompts (?), sensorimotor sequences (Fu et al., 2024), or geometric demonstration representations (Vosylius and Johns, 2024). RICL explicitly develops this capability in pretrained vision–language–action models through dedicated finetuning (?). Other methods learn to use human visual demonstrations as task context (?Patel et al., 2026; Zhou et al., 2026), while GEN-1.5 and S1 report adaptation from sensorimotor or video prompts (Generalist Team, 2026; Skild AI, 2026). Beyond fixed-weight conditioning, RoboTTT incorporates long-context visual experience through gradient-updated fast weights (Jiang et al., 2026). These studies establish diverse mechanisms for demonstration-driven adaptation in embodied models. We investigate such adaptation in an off-the-shelf general-purpose multimodal agent, using visual demonstrations and interaction history without additional robot-specific training or test-time parameter updates.

General-purpose agents for robot control. Language and vision-language models have been introduced at multiple levels of the robotic control stack. SayCan grounds language-model predictions in the affordances of

a library of pretrained skills (?); Code as Policies synthesizes executable programs over predefined perception and control APIs (?); and VoxPoser translates language and visual reasoning into spatial objectives for downstream motion planning (?). These systems demonstrate that general-purpose models can reason about embodied tasks, but physical execution remains delegated to a separate skill library, program, planner, or controller. RoboPrompt moves closer to action-level control by converting visual keyframes, object poses, and expert end-effector actions into textual prompts from which an off-the-shelf LLM predicts subsequent actions (Yin et al., 2025). Its interface, however, relies on structured state representations and expert action annotations rather than raw visual task specifications. Prior systems therefore use general-purpose models primarily to specify, compose, or guide behavior executed by a separate control system. We instead place GPT-6 directly in the closed loop, condition it on raw visual keyframes without expert action labels, and investigate whether a general agent can itself serve as both the embodied policy and the in-context learner.

3 Method

GPT-POLICY connects a fixed general VLM to robot tools through a context-to-action interface (Figure 2). We describe the task formulation, context construction, and closed-loop execution separately.

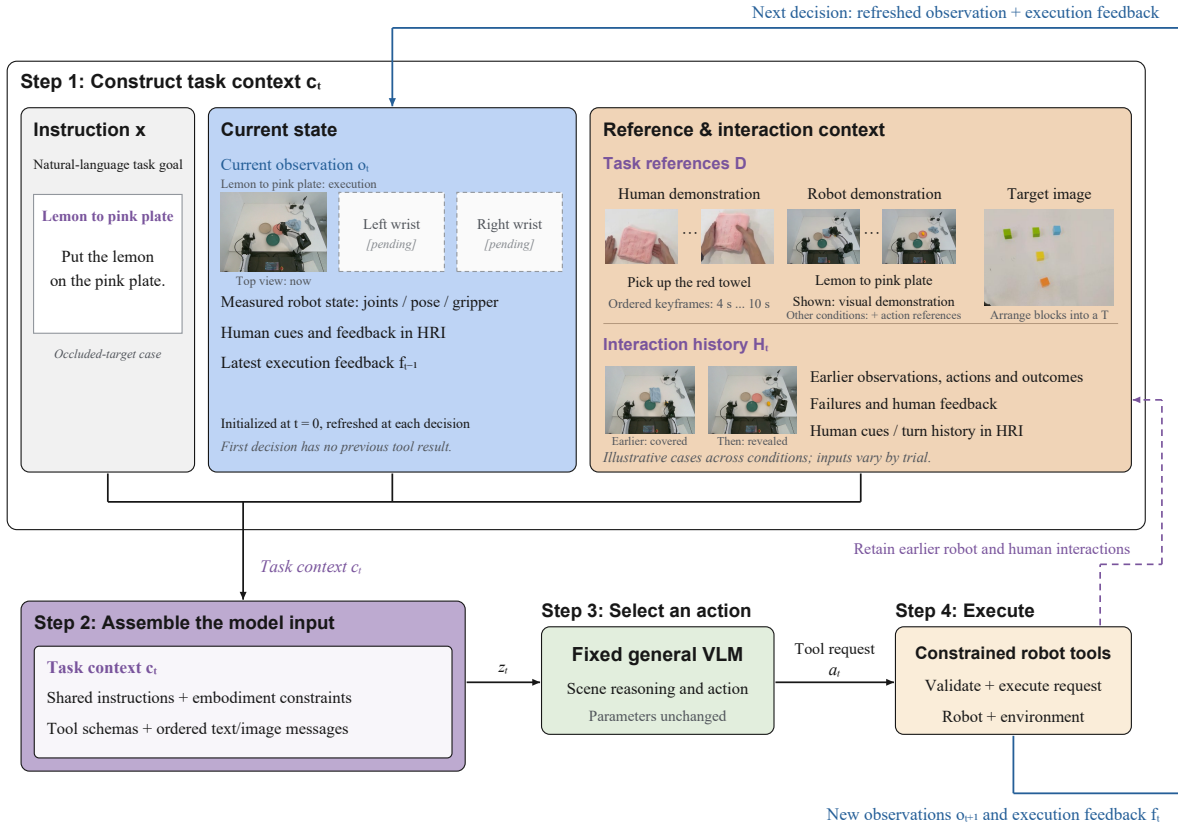


Figure 2 Overview of the context-to-action framework. Task instruction, current state, and reference and interaction context form the task context. Model-input assembly combines this context with shared instructions and tool schemas. A fixed VLM selects tool requests, and constrained robot tools return observations and execution feedback for the next decision. The images illustrate different experimental conditions.

3.1 Problem Formulation and Context Inputs

Given an instruction x and current observations o_t , the agent uses external examples D and preceding interaction history H_t to select an executable action. Examples may contain human or robot demonstrations, optional action references, or a goal image; history contains earlier observations, actions, outcomes, and human

feedback. The context condition m defines the permitted information:

$$c_t = \mathcal{C}(D, H_t; m). \quad (1)$$

The model parameters θ remain fixed, with no gradient updates or persistent task-specific parameter changes. Current observations and the latest execution feedback are available independently of retained history. Adaptation is evaluated from a new initial state: the model must ground the contextual information in the current scene and generate new tool requests.

3.2 Context Construction and Management

Shared instructions and action rationale. A fixed system prompt defines the robot-control protocol, including observation checks, call budgets, stopping rules, and platform-specific coordinate and motion conventions. Each motion includes a one- or two-sentence **note** describing the current observation and the reason for the chosen motion. This is an exposed action rationale, not a prescribed chain-of-thought trace; reasoning effort is a separate backend setting. Codex returns one tool name and JSON-encoded arguments for host-side validation (Figure 3). Shared instructions remain fixed within matched comparisons, while optional prior-attempt notes are recorded as condition-specific inputs.

Demonstration and observation formats. A versioned demonstration bundle contains source and coverage metadata, time-ordered keyframe records, and camera-labeled images. State and recorded actions are optional references; unknown values use an empty state mapping and a null action. YAM demonstration preparation defaults to the top view, with up to three named views supported per keyframe. In contrast, the reference YAM runner supplies three live RGB views—top, left wrist, and right wrist—resized to the configured resolution, with depth input disabled. Camera labels precede image blocks; robot state and execution feedback are serialized as text. Coordinate conventions are provided, but the default serializer does not attach camera intrinsic or extrinsic matrices. Separate calibration files support geometric utilities and are not automatically included in the model input.

Assembly and input constraints. Shared instructions, the optional demonstration, and the current goal initialize the conversation; live observations and tool outcomes then accumulate in a trial-specific session. Historical examples are explicitly separated from current observations and cannot establish current-trial success. The loader checks increasing nonnegative timestamps, finite state values, 1–24 keyframes, 1–3 PNG/JPEG views per frame, and a total limit of 48 images and 32 MiB. Frame selection and visual-action alignment are performed offline. Visual-only conditions use neutral annotation text and omit procedural descriptions and recorded actions; action-channel comparisons preserve images and non-action text. Preparation excludes scored-trial future information. Codex manages retained context server-side, so explicit history ablations require reconstructed inputs and fresh sessions (Appendix A.3).

3.3 Action Generation and Closed-Loop Execution

The agent selects a tool request a_t using the current scene, context, and latest feedback f_{t-1} :

$$a_t = \pi_\theta(x, c_t, o_t, f_{t-1}), \quad (o_{t+1}, f_t) = \text{Execute}(a_t). \quad (2)$$

No previous tool result is present at the first decision. A motion chunk executes or is interrupted before the next model decision; fresh observations and the execution result then support replanning. A termination request is recorded separately from physical success.

Pose requests and interpolation. After arm selection, the agent issues one target through **move_to** or two to eight through **move_eef_chunk**. An absolute-pose target consists of position $p_j \in \mathbb{R}^3$, orientation $R_j \in SO(3)$, and gripper command g_j . Positions are expressed in metres and orientations are transmitted as

Persistent instructions and example context Robot protocol + embodiment notes + tool schemas Historical demonstration: source, demonstrator, coverage, coordinate frame, outcome Keyframe i : <code>t_s</code> , <code>stage</code> , <code>observation</code> , <code>result</code> , <code>state</code> , optional <code>action</code> Camera label + image; repeat for retained keyframes and views Current goal: x
Live observation at decision t (reference YAM runner) Current observation. instruction + <code>state[eef_pose]</code> / <code>state[joint_pos]</code> Controller execution feedback + optional operator feedback camera 'top_cam': [RGB] 'left_cam': [RGB] 'right_cam': [RGB] <i>No depth or camera calibration matrices are attached by default.</i>
Structured model output (illustrative motion request) <pre>{ "name": "move_to", "arguments": "<JSON-encoded object>" }</pre> Decoded arguments: targets : pose fields; note : observed evidence + motion rationale → host validation and execution → next observation and tool result

Figure 3 Schematic input and output format, not an execution trace. Text and image blocks are interleaved. The absolute-pose adapter labels pose feedback as `eef_pose`; the legacy adapter uses `eef_state`. A goal image can supply the target configuration without a temporal demonstration. The short action note is distinct from any unobserved internal reasoning.

unit $xyzw$ quaternions in each arm’s base frame. For adjacent targets, the adapter constructs

$$T_j(s) = \begin{bmatrix} R_j(s) & p_j(s) \\ 0^\top & 1 \end{bmatrix}, \quad \begin{aligned} p_j(s) &= (1-s)p_j + sp_{j+1}, \\ R_j(s) &= R_j \exp(s \log(R_j^\top R_{j+1})), \end{aligned} \quad s \in [0, 1]. \quad (3)$$

The rotational path is implemented by shortest-arc quaternion SLERP. A local Ruckig profile supplies $s(t)$, with rest at pose junctions. The legacy coordinate adapter instead plans directly in its end-effector coordinates; the two modes are recorded separately. Omitted fields inherit the preceding target, initialized from the last acknowledged command, while interpolation normally starts from measured pose. Durations follow configured motion limits. A new gripper command is allowed only in the final target and is interpolated during the final segment; a separate gripper-only target permits approach before closure.

Inverse kinematics and path validation. For each sampled target $\hat{T}_k = (\hat{R}_k, \hat{p}_k)$, the absolute-pose adapter calls the platform IK solver using the preceding joint reference as its seed:

$$q_k = \text{IK}(\hat{T}_k; q_{k-1}), \quad e_{p,k} = p(q_k) - \hat{p}_k, \quad e_{R,k} = \text{Log}(\hat{R}_k^\top R(q_k))^\vee, \quad (4)$$

where $(R(q), p(q))$ is forward kinematics and $(\cdot)^\vee$ maps a skew-symmetric matrix to its rotation vector. The solver must converge, and each resulting joint reference must pass

$$\begin{aligned} \|e_{p,k}\|_2 &\leq \epsilon_p, & \|e_{R,k}\|_2 &\leq \epsilon_R, \\ q_{\min} &\leq q_k \leq q_{\max}, & \|q_k - q_{k-1}\|_\infty &\leq \Delta_q. \end{aligned} \quad (5)$$

The implementation rechecks pose residuals after applying effective joint bounds and rejects failed checks before dispatch. These conditions describe solver acceptance, not measured tracking. Geometric collision checking additionally requires a calibrated scene; the explicitly configured visual-test mode omits that check.

Execution feedback. Runtime tracking and hardware guards remain active during execution. Rejected targets and pose-tracking errors return feedback for replanning; a processed waypoint does not establish measured arrival or task success. The tools and checks remain fixed within matched comparisons. Platform configurations, including the separately specified mobile interface, are recorded in Appendix C.

4 Experiments

This section evaluates whether a general-purpose agent can adapt to tasks during closed-loop robotic-arm operation by using different forms of context. The experiments are designed to answer three questions:

1. What information does each form of context provide?
2. Can this information improve task completion rates and reduce the number of decisions?
3. Which forms of context remain effective for fine contact-rich manipulation, deformable-object handling, and human-robot collaboration?

4.1 Unified Experimental Setup

Each episode starts from a reset scene and ends when the task succeeds, the execution budget is exhausted, or a safety termination condition is triggered. An episode is counted as successful only when the final scene satisfies the task-specific geometric and semantic success criteria. All conditions use the same success criteria and termination rules. Each experiment is repeated three times.

The main metrics are:

- **Success Rate (S/T):** S denotes the number of successful trials and T denotes the total number of trials; the metric is reported as S/T .
- **Decisions:** The number of decisions generated by the agent in one episode. Under the task-specific counting convention, each generated action target or action block counts as one decision.
- **Execution Time:** The total time required to complete the task.

The experimental results are summarized in Table 1.

Model	ICL Condition	Task 1				Task 2			
		Task Name	S/T	Decisions	Time (min)	Task Name	S/T	Decisions	Time (min)
Human Video Demonstration									
GPT-6 Astra	w/o Human Video	Pick Red Towel	0/3	96.3	24.6	Remove Glue Cap	3/3	49.3	12.2
GPT-6 Astra	Human Video	Pick Red Towel	2/3	76.7	18.9	Remove Glue Cap	3/3	65.7	16.9
Robot Visual Demonstration									
GPT-6 Astra	Robot Video	Unscrew Bottle Cap	–	–	–	Insert Plug	–	–	–
GPT-6 Astra	Robot Video + Action	Unscrew Bottle Cap	–	–	–	Insert Plug	–	–	–
Target Image									
GPT-6 Astra	Target Image	Arrange T Shape	3/3	59.3	13.3	Arrange Fruit	3/3	49.0	12.4
Self Interaction History									
GPT-6 Astra	Self History	Lemon To Pink Plate	3/3	35.0	8.1	Movable Exploration	3/3	40.33	25.53
Online Human-Robot Interaction									
GPT-6 Astra	Human-Robot Interaction	Tic-Tac-Toe	3/3	69.7	13.6	Pointed Fruit Pickup	3/3	67.3	15.0

Table 1 Results Across Different Context Conditions. Here, S/T denotes the number of successful trials divided by the total number of trials.

4.2 Human Video Demonstration-Driven Context Learning

This experiment examines whether a general-purpose agent can understand a human operation video as a task specification. The agent receives one human video demonstration, a natural-language task instruction, and real-time multi-view observations of the robotic arm. As a control, we include a condition that receives the same textual instruction and observations without the video demonstration.

The two tasks in this section are picking up the red towel with the right hand (Pick Red Towel) and removing the cap from a glue stick (Remove Glue Cap). Table 1 reports the success rate, number of decisions, and execution time for these tasks under the corresponding context conditions.

The results show that, after receiving Human Video Demonstration, GPT-6 Astra and Claude-Fable 5.1 can interpret the human operation video as a task specification and guide the robotic arm to follow the demonstrated procedure. By comparison, success is lower without the human video demonstration, even when

the textual instruction describes the task in detail. This indicates that a human video demonstration provides useful contextual support beyond a detailed textual instruction.

4.3 Robot Visual Demonstration and Action Reference-Driven Context Learning

This experiment examines whether a general-purpose agent can use Robot Visual Demonstration and the corresponding Action Reference to perform fine-grained manipulation. The agent receives a robot demonstration video, the action reference synchronized with the video, a natural-language task instruction, and real-time multi-view observations of the robotic arm. As controls, we include a condition with no demonstration and a condition that provides the robot visual demonstration without the action reference. The two demonstration conditions use the same video; the action reference contains synchronized end-effector poses, gripper states, and action commands.

The two tasks in this section are Unscrew Bottle Cap, which requires unscrewing the bottle cap and leaving the bottle standing securely, and Insert Plug, which requires inserting the plug into the corresponding socket. Table 1 reports the success rate, number of decisions, and execution time for these tasks.

The results show that GPT-6 Astra and Claude-Fable 5.1 can use the robot visual demonstration and action reference to transfer the demonstrated procedure into the current scene and complete Unscrew Bottle Cap and Insert Plug. This indicates that robot visual demonstrations and action references provide useful contextual support for fine-grained manipulation.

4.4 Target Image or Result Demonstration-Driven Context Learning

This experiment examines whether a general-purpose agent can interpret a Target Image as a requirement on the final task state. The agent receives a Target Image, a natural-language task instruction, and real-time multi-view observations of the robotic arm. As a control, we include a condition that receives the same textual instruction and observations without the target image.

The two tasks in this section are Arrange T Shape, in which objects are arranged into a T shape, and Watercolor. Watercolor is marked as pending procurement in the experiment summary; the corresponding task column is retained in the table and its results remain to be completed. Table 1 reports the success rate, number of decisions, and execution time for these tasks.

For the completed Arrange T Shape task, the results show that adding a target image enables GPT-6 Astra and Claude-Fable 5.1 to understand object relations and the target layout and to guide the robotic arm to arrange the objects accordingly. This indicates that a target image provides clear spatial constraints, although converting those constraints into robotic behavior still depends on the model’s ability to understand and use visual context.

4.5 Self Interaction History-Based Context Learning

This experiment examines whether a general-purpose agent can use its own interaction history to explore and adapt subsequent actions. The agent receives a natural-language task instruction, current multi-view observations, and observations, actions, execution results, and failure feedback generated earlier in the same task. This history is retained as context, and the model parameters are not updated during execution.

The two tasks in this section are Lemon To Pink Plate, in which the robot searches for the target and places the lemon on the pink plate, and Movable Exploration, in which the robot searches for a target by changing its observation position or moving an occluding object. Table 1 reports the success rate, number of decisions, and execution time for these tasks.

The results show that GPT-6 Astra and Claude-Fable 5.1 can use previous observations, actions, and execution results to adjust subsequent search and manipulation and complete Lemon To Pink Plate and Movable Exploration. This indicates that self interaction history can support later decisions, while successful execution still requires the model to apply prior interaction experience to the current decision.

4.6 Online Human-Robot Interaction-Driven Context Learning

This experiment examines whether a general-purpose agent can understand human interaction information generated during execution and use it to adapt the robotic arm’s behavior. The agent receives a natural-language task instruction, real-time multi-view observations, and Online Human-Robot Interaction Context, such as pointing actions or move history. We compare the models under the same interaction condition and examine whether they can update target selection and action timing in response to human actions.

The two tasks in this section are Tic-Tac-Toe, played through interaction with a human, and Pointed Fruit Pickup, in which the robot grasps the fruit indicated by the human. Both tasks use Online Human-Robot Interaction Context. Table 1 reports the success rate, number of decisions, and execution time for these tasks.

The results show that GPT-6 Astra and Claude-Fable 5.1 can use online human interaction information to select the pointed fruit and update subsequent actions based on the human’s move, guiding the robotic arm through the corresponding interactive tasks. This indicates that online human interaction can support target selection and turn-taking, while the ability to convert interaction information into behavior still differs across models.

5 Discussion

Information supplied by context. The cases motivate asking which missing information changes a decision. Goal images specify an outcome, demonstrations reveal an operation, and history can preserve information no longer visible. This suggests testing transfer by changing the layout, goal relation, or completed substeps while preserving the relevant contextual information. Behavior that follows these changes would provide stronger evidence of adaptation than reproducing a demonstrated sequence. The current comparisons motivate this test but do not yet establish its reliability.

Reasoning and execution. Correct task interpretation can coexist with failed physical execution, as the bottle traces illustrate. More context may help identify the next subgoal without resolving contact, calibration, or tracking errors. General agents could therefore handle contextual interpretation while specialized robot controllers handle frequent physical corrections; whether this division improves performance requires evaluation. Mobile exploration extends the same issue to deciding where to observe and preserving that information until it becomes relevant to an action.

Limitations and safety. The evidence comes from small run series and selected historical comparisons, with incomplete baselines and controlled ablations. Differences in prompts, initial states, and verification limit causal conclusions, while broader tests of contact-rich and mobile tasks are needed to establish reliability and transfer. Unobserved pretraining also limits claims of novel skill acquisition. Safety depends on calibration, execution guards, and timely feedback, which cannot cover all contacts or changes in the surroundings. The current results therefore do not establish safe autonomous deployment; independent monitoring and human intervention remain necessary.

6 Conclusion

We study how fixed general agents use demonstrations, goal images, and interaction experience in robotic tasks without parameter updates. GPT-POLICY provides a shared context-to-action interface for examining these inputs. The recorded behaviors illustrate goal grounding, changes in operation order, and online coordination, while contact execution and outcome verification remain distinct challenges. The task-level model comparisons and reserved ablations provide a basis for testing when context improves success, efficiency, and recovery across manipulation and mobile exploration.

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Appendix

A Reporting and Reproducibility Protocol

A.1 Trial accounting and efficiency

All attempted trials, including failures, timeouts, rejected motions, and interventions, belong in the trial ledger. A completed model response is not a completed robot motion, and a termination message is not independent verification of success. Each trial records the model identifier, reasoning setting, prompt and tool-schema versions, robot and camera configuration, context inputs, initial state, budgets, stopping reason, and outcome evidence. Controlled condition pairs use matched scene variables and randomized order. The current repeated-trial plan uses three attempts per cell; counts must accompany percentages, and small samples should not be presented as precise estimates of reliability.

An action decision is a model invocation that requests an action or action chunk, regardless of acceptance. Arm selection, observation-only calls, and termination are excluded from this action count and retained in a separate total request count R . A chunk with several waypoints is one action decision; calls containing several tools require an explicit grouping rule in the transcript parser. A command that only refreshes observations without a physical action is labeled observation-only. The mean action-decision count over successful trials is

$$D_{\text{succ}} = \frac{\sum_{i=1}^n y_i D_i}{\sum_{i=1}^n y_i}, \quad (6)$$

where y_i indicates verified task success. If no trial succeeds, D_{succ} is undefined. The historical summaries provide R rather than consistently annotated D_i , so the main table labels their successful-trial mean as R_{succ} . These quantities must not be silently interchanged. Timeouts remain in total requests and failure accounting even when they return no action.

Runtime includes initialization, model inference, robot motion, human waiting where applicable, and cleanup. Model latency and motion time are reported separately when available. Neither a short failed run nor a selected successful video is a basis for an overall efficiency claim. Per-task records accompany any family aggregation; model comparisons use the same task coverage and weighting. Costs and token counts in the supplied summaries are estimates and are not treated as audited billing measurements.

A.2 Input and outcome reconciliation

The current table transcribes run summaries supplied for the paper and does not imply an independent audit of the remote execution logs. The following checks are required before the standardized comparison is finalized.

- The red-towel series reports successes at 61 and 56 requests and a failure at 113. Its successful-trial mean is 58.5; 76.7 is the mean over all three attempts. A standardized rerun is planned.
- Glue-stick cap removal reports 98, 43, and 56 requests, all successful. The corresponding mean is 65.7. The self-history lemon series reports 31, 36, and 39, giving 35.3 rather than the rounded value 35 in the source summary. Pointing-guided selection reports 50, 85, and 67, giving 67.3.
- T-arrangement records list image-conditioned runs at 45, 67, and 66 requests. The no-image row repeats these numbers without an independent run list. Verify the actual image inputs and unique run identifiers before filling that baseline. Input-standardized repeats are also planned.
- The English pointing instruction waits for an OK gesture to terminate, while the Chinese description specifies two placements. The tic-tac-toe English instruction uses green pieces, while the Chinese and historical descriptions use yellow. Each scored run must be adjudicated against its actual English prompt; color, cue sequence, and stopping rule must be fixed across conditions.
- Historical bottle trials use different no-demonstration wording, and the two demonstrated inputs contain visual stage text. The plug inputs share image timestamps but differ in stage wording. These trials are

case analyses, not clean action-channel ablations.

- Bottle completion is a model claim without independent acceptance in the supplied record. Plug insertion has a recorded final-image review but no electrical test or operator acceptance. A readable video is not evidence of a complete synchronized robot-state trace.

The formal roster excludes gesture dance and tissue-box self-correction, which are marked outside the paper in the current task sheet. Watercolor, mat placement, and other exploratory ideas have no current repeated-trial slots. Mobile exploration remains in the roster while its target, action, and platform configuration are resolved. Historical examples and standardized repeats retain separate identifiers and are never counted twice.

A.3 Planned Ablation Matrix

TODO: These ablations are optional, subject to available experimental time. All results are pending; finalize the subset and run matched trials before presenting any ablation conclusions.

The ablations isolate the information needed for the behavioral effects above. Tables 2 and 3 specify individual settings, task subsets, and result fields. Each row specifies one primary task and setting, to be evaluated on three paired initial layouts. The remaining tasks provide subsequent replication. The initial ablation model is GPT-6-Astra with medium reasoning, except when reasoning effort is the factor under test; replication on other models follows the same protocol. The bold settings are proposed comparison references, not claims about the configuration of historical runs. All result cells remain pending.

Context content and representation. The supplied bundle loader permits at most 24 keyframes; the largest setting respects this limit. The frame-count study changes only how many uniformly spaced frames are retained from the same source video and view. The action-reference study holds video frames and non-action text fixed; time-shuffled references test whether alignment matters beyond the presence of action values. The image-resolution study preserves crop, aspect ratio, and the target criterion while reducing the native resolution. The prompt study compares a goal alone, a written procedure, and the corresponding human video; the two procedural conditions describe the same operation, and no procedural text is appended to the video condition. This separates the contribution of additional task information from how it is represented.

Table 2 Context ablations, with one row per setting. Each row will report s/n , successful-trial action decisions D_{succ} , and successful-trial runtime T_{succ} in seconds. Dashes denote pending results.

Factor	Task	Setting	s/n	D_{succ}	T_{succ}
Video frames	Towel pickup	4 frames	—	—	—
		8 frames	—	—	—
		14 frames	—	—	—
		24 frames	—	—	—
Action reference	Plug insertion	Visual only	—	—	—
		Visual + aligned actions	—	—	—
		Visual + shuffled actions	—	—	—
Goal resolution	T arrangement	0.25× native size	—	—	—
		0.5× native size	—	—	—
		Native size	—	—	—
Prompt content	Glue cap removal	Goal only	—	—	—
		Goal + written procedure	—	—	—
		Goal + human video	—	—	—

History and execution. History length counts preceding observation–action–outcome cycles. The current observation and latest execution feedback remain available at every length, including zero; the scored current scene is matched across the history conditions. For Codex, this planned test requires a fresh session with an explicitly reconstructed input at each scored decision: its native thread cannot be truncated, and

`image_horizon` is unsupported. This infrastructure is not yet established by the supplied code. The test is planned first on target search and then extended to mobile exploration after its target and movement interface are fixed. The one-target condition uses `move_to`; the other conditions also permit `move_eef_chunk`. Chunk size is the maximum number of end-effector waypoints permitted in a motion request; the controller, motion limits, model-request cap, and task time limit remain fixed. Reasoning effort is varied with the same inputs and tools. Request counts, rejected targets, and task failures accompany successful-trial efficiency so that longer chunks or early failure cannot create an apparent gain by accounting alone.

Table 3 History, action-chunk, and reasoning ablations. Each task and setting has independent result fields under the same accounting convention as Table 2. Mobile exploration is an extension of the history test pending task specification.

Factor	Task	Setting	s/n	D_{succ}	T_{succ}
Retained history	Target search	0 prior cycles	–	–	–
		Last 2 cycles	–	–	–
		Last 4 cycles	–	–	–
		Full episode history	–	–	–
Chunk size	Plug insertion	1 target (<code>move_to</code>)	–	–	–
		Up to 4 waypoints	–	–	–
		Up to 8 waypoints	–	–	–
Reasoning effort	Target search	Low	–	–	–
		Medium	–	–	–
		High	–	–	–

Temporal shuffling of video, semantic mismatch, keyframe selection method, and cross-model replication provide follow-up controls once the primary settings are measured. Appendix A specifies the input ledger and trial accounting. Raw historical request counts are not substituted for action-only decisions in these ablation tables.

A.4 Additional Diagnostic Comparisons

The following comparisons specify the evidence needed to interpret and extend the planned ablations.

Table 4 Context interventions and the behavior they test.

Family	Critical comparison	Measurements beyond completion
F1	Ordered / shuffled / mismatched human video, with no procedural text	Operation order and grasp-method transfer
F2	Identical frames and text, with / without aligned actions	Contact-stage completion, alignment corrections, rejected targets
F3	Matched / missing / mismatched target images	Spatial relations and geometric agreement
F4	Same current view with retained / truncated / removed history	Discoveries, revisits, repeated errors, and recovery
F5	Current cue alone / current cue plus interaction history	Cue following, turn timing, response to correction

Video-budget studies can vary frame count (for example, 4, 8, 14, and 24) or selection method (uniform, event-based, or model-selected), one at a time. Camera view and goal-image resolution are separate factors. Context count, image tokens, text tokens, action annotations, and history retention are recorded explicitly. Equal token counts do not guarantee equal task information across modalities. Reasoning-budget and chunk-size studies retain the same context and controller limits, and report request count, action count, physical execution, and latency together. Safety checks are not disabled as an ablation.

For mobile exploration, both matched-view history tests and full-episode tests are needed: the latter allow different actions to create different future observations. Base travel or viewpoint changes, target discovery,

and the final manipulation are logged separately. For human interaction, retain the current cue in the reactive baseline. Learning a new convention requires histories that establish that convention; success on a familiar game or an explicit pointing instruction is not sufficient by itself.

A.5 Extended models and per-trial reporting

GLM-5.3-flash and KIMI-3 are planned model extensions. The $\pi_{0.5}$ reference uses no target-task finetuning and requires a compatible observation and action interface and comparable task coverage. Its success rate is reserved in the main grid; unsupported task cells should be marked not applicable once compatibility is established. Robot-policy control steps are not equated with general-agent model requests.

Table 5 Reserved per-task efficiency fields. All-attempt statistics accompany the successful-trial action count.

Model / task / condition	D_{succ}	Total requests	Runtime	Rejections / interventions
Matched baseline	—	—	—	—
Matched context	—	—	—	—
Ablation condition	—	—	—	—

B Episode Metadata

The expert demonstration episode is `episode_0908_055812_165190`. The no-context policy episode is `20260907T222432Z-no-demo-60-eval`; the visual-context episode is `20260907T233301Z-icl1-top-60-camera1s-eval`. Decision counts were cross-checked against model transcripts and decision-latency entries. The videos are 640×480 at 30 frames per second, with small duration differences across views.

C Experimental Configurations

This table distinguishes the current repeated-evaluation plan from historical case settings. Unspecified values remain open protocol fields and are not inferred from a different platform or run.

D Task Success Criteria

These predicates define the intended standardized evaluation for the current task roster. Numeric tolerances, observation windows, target instances, and acceptance rules must be frozen before scored repeats. Historical completion labels are retained as recorded and are not retroactively treated as having passed these proposed criteria.

A familiar-game win or response to a currently visible pointing cue measures task competence and coordination. A separate history intervention is required to establish reliance on previous experience. The external-demonstration occluded-target case is not pooled with self-history lemon trials. All conditions preserve current observations and operational feedback, and all motion rejections and human interventions remain in the trial record.

Table 6 Configuration ledger for the repeated comparison and archived examples.

Configuration	Recorded value or pending field
Platforms	Fangzhou 247/248 and YAM; exact hardware revisions, calibration, and per-task allocation pending
Primary models	GPT-6 and Claude-Fable 5.1; deployment identifiers and access dates recorded per trial
Historical model setting	GPT-6-Astra / medium for the archived task comparisons
Extended models	GLM-5.3-flash and KIMI-3; no comparison results yet
Repetition plan	Three attempts per model-task-condition cell; history baselines are additional reserved comparisons
Task language	English in the repeated evaluation; historical prompts retained separately
Scenes	Manual reset between attempts; paired layout variables and randomized order to be frozen for controlled trials
Cameras	Historical YAM records use top, left-wrist, and right-wrist views; confirm view mapping and calibration for each platform
Reference YAM input	Three live RGB views (top, left wrist, right wrist), robot state, and execution feedback; depth disabled in the inspected runner
Image resolution	Inspected example configs use 224×224 live images; verify actual live and demonstration resolutions per run rather than inheriting this default
Model-visible geometry	Coordinate conventions are supplied; camera intrinsic/extrinsic matrices are not attached by the default serializer. Record any additional calibration text or geometric input explicitly
Prompting controls	Record shared system/embodiment instructions, tool schemas, reasoning setting, and any imported prior-attempt notes; hold them fixed within each context comparison
Historical control interface	AnyRobo, absolute quaternion end-effector targets, inverse kinematics and Ruckig timing in the archived bottle/plug runs
Historical bottle/plug budgets	100 model requests; 1800 s task limit; 60 s decision timeout; maximum speed fraction 0.25
Repeated-evaluation limits	Exact request, action, runtime, motion, and intervention budgets to be verified from each configuration; historical defaults are not assumed
Mobile task	Target object, terminal operation, occlusion layout, base or viewpoint-control tools, travel limits, and scene reset pending
Input versions	Freeze video/image hashes, actual frame/view counts and timestamps, annotation text, action coordinates and units, prompt text, and tool schema
Outcome verification	Record operator or image-based adjudication separately from model termination; preserve failed and interrupted trials

Table 7 Success criteria for demonstration and goal-image tasks.

Task	Full-task predicate	Additional annotation
Red-towel pickup	Lift and hold the designated towel with the right arm for a fixed interval without dropping it	Score the demonstrated grasping method separately from pickup success
Glue-stick cap removal	Fully separate the cap while controlling the glue stick	Stabilization, cap engagement, separation, final verification
Bottle-cap unscrewing	Detach the cap and leave the bottle standing securely	Engagement, rotation, detachment; independently verify completion
Plug insertion	Mechanically seat the plug in the designated socket to a fixed depth or visible seating criterion	Alignment and seating; electrical operation is outside this criterion
T-shaped arrangement	Place the specified blocks in the required color and spatial relations within fixed tolerances	Report spacing and position error; use the same target criterion across conditions

Table 8 Success criteria for exploration and human interaction. Task completion and evidence of history use are separate outcomes.

Task	Full-task predicate	Additional annotation
Target search and placement	Locate the pink plate and release the lemon onto it from the scored starting state	Search, discovery, placement, final verification
Mobile exploration	Discover the predefined target by permitted movement and complete the specified target operation within budget	Target and terminal action pending; record travel, views, discoveries, revisits, and manipulation separately
Tic-tac-toe	Complete a legal alternating game under a fixed opponent and cue protocol	Report win/draw/loss, illegal moves, turn timing, and piece colors separately
Pointing-guided selection	Select and place the indicated fruit for each cue and terminate under the fixed English stopping rule	Selection accuracy, placements, response to changed cues, and termination; resolve OK-gesture versus fixed-count wording